

TrES-5b

R-0716

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-5_251001 · created 2026-07-27T23:29:45+00:00

BENCHMARK: INACCURATE

Results

Mid-Transit Time (Tmid)	2460949.8059 +/- 0.0039 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.183 +/- 0.017
Transit depth (Rp/Rs)^2	3.34 +/- 0.61 [%]
Orbital Inclination (inc)	89.56 +/- 1.65
Airmass coefficient 1 (a1)	1.046 +/- 0.03
Airmass coefficient 2 (a2)	-0.023 +/- 0.017
Scatter in the residuals of the lightcurve fit is	2.91 %
Best Comparison Star	#3 - [343, 95]
Optimal Aperture	3.06
Optimal Annulus	13.51
Transit Duration (day)	0.0895 +/- 0.0021

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.183 $\pm$ 0.017 vs 0.142 (deviation 2.13 σ)
Duration fit vs published	0.0895 d vs 0.0758 d (deviation 5.74 σ)
Mid-time O-C	9.5 min
Depth SNR	9.5
Residual scatter	2.91 %

Light curve

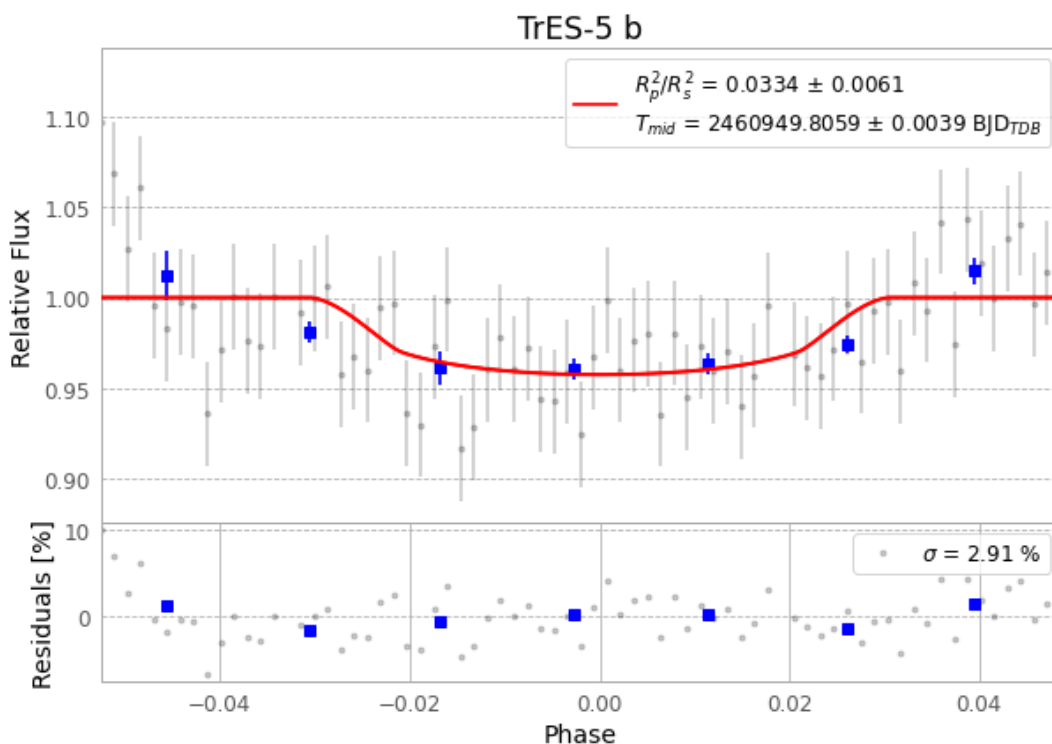


Figure 1 — FinalLightCurve_TrES-5 b_2025-09-30.png (SHA-256 608283c44c88d7dc...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [258, 339], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 8b3c93f0a7834247...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ead60b9

Data provenance

73 FITS frames (48 MB), TRES-5251001052420.FITS → TRES-5251001090015.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-5-251001.log (511fc875a07d...), FinalLightCurve_TrES-5 b_2025-09-30.png (608283c44c88...), FinalLightCurve_TrES-5 b_2025-09-30.csv (bab43d9facb8...)

Compute resources

Wall time 115.1 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-27 23:29Z.